

Smart Classroom Automation System Using STM32

"Smarter Classroom, Better Comfort, Efficient Energy Use"

A proposed closed-loop embedded automation architecture that integrates multi-sensor environmental acquisition with ARM Cortex-M processing to autonomously govern lighting, thermal comfort, and energy preservation in institutional learning spaces.



CORE MCU

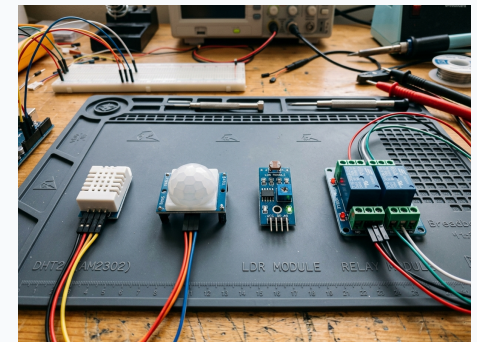
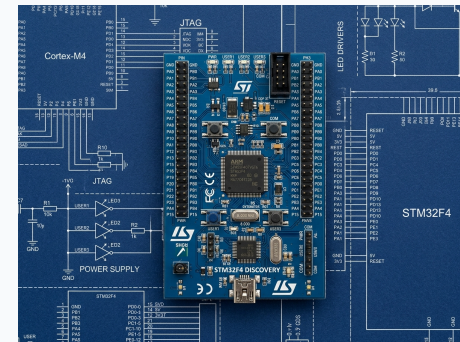
STM32 (ARM Cortex)

IOT GATEWAY

ESP32 Wi-Fi

TARGET OUTPUT

Relay & PWM Cooling



Introduction & Context

The Need for Intelligent Energy Management in Academic Spaces

Academic classrooms and lecture halls present dynamic physical environments characterized by erratic schedules, shifting thermal loads, and variable outdoor daylight throughout the day.

- **Heavy Energy Footprint:**

High-bay fluorescent/LED lighting and ceiling fans run continuously regardless of occupancy.

- **Static Manual Control Dependence:**

Human occupants frequently forget to switch off loads upon dismissal, causing waste.

- **Dynamic Environmental Fluctuation:**

Solar daylight streaming through perimeter windows often renders artificial lighting redundant.



CORE HYPOTHESIS

Automated closed-loop regulation achieves up to 30–45% energy savings while simultaneously improving student learning comfort.

Problem Statement

Quantifying Inefficiencies in Conventional Institutional Infrastructure

Daylight Ignorance

Artificial lights burn at 100% capacity even when outdoor sunlight provides over 600 lux across student desktops through exterior windows.

Zero-Occupancy Waste

Ceiling fans and lights continue spinning and glowing during empty periods, between lecture slots, and after evening closures.

Fixed-Speed Discomfort

Fans operate on primitive binary or manual rotary step switches that do not adapt dynamically to fluctuating thermal loads.

Absence of Telemetry

Campus facility administrators possess no real-time telemetry regarding room thermal compliance, presence, or status.

Cumulative Utility Costs

Across campuses with 50+ lecture halls, manual negligence translates to thousands of kilowatt-hours wasted annually, straining institution budgets.

Project Objectives

Clear, Measurable Engineering Deliverables of the Proposed System

[01] Environmental Sensory Acquisition

Accurately sample ambient temperature, humidity, and daylight illuminance via digital & analog interfaces.

[02] Passive Human Occupancy Detection

Detect human entrance, movement, and physical occupancy vectors without invasive biometric tracking.

[03] Dual-Condition Smart Lighting

Automate high-voltage lighting circuits via solid-state relays only when both occupancy and daylight deficit occur.

[04] Proportional PWM Thermal Cooling

Dynamically modulate DC fan velocity across distinct thermal brackets (30%, 60%, 100%) based on ambient heat load.

[05] Zero-Waste Inactivity Latch

Automatically de-energize all lighting and cooling circuits after confirmed absence to eliminate overnight parasitic power draw.

[06] Wireless Telemetry & IoT Gateway

Interface STM32 with ESP32 over UART to transmit continuous telemetry packets over Wi-Fi.

[07] Remote Operator Override Capability

Enable facility managers to view live environmental trends and manually force actuators when specialized events require it.

Proposed System Overview

High-Level Input-Process-Output Automation Topology

1. SENSORY INPUTS

- **DHT22 Sensor**

Digital Single-Bus Temp & Humidity

- **LDR Module**

Analog Lux Illuminance (12-bit ADC)

- **PIR Motion**

Passive Infrared GPIO EXTI Interrupt

2. MAIN CONTROLLER STM32 (ARM Cortex-M)

- ✓ Hardware Timer PWM (TIM2_CH1)
- ✓ Continuous multi-variable evaluation
- ✓ Inactivity countdown latch state machine
- ✓ USART2 packet assembly & serialization
- ✓ Deterministic sub-50ms execution loop

3. ACTUATORS & GATEWAY

- **Opto Relay**

230V AC Lighting Troffer Control

- **MOSFET Driver**

PWM Speed Control for DC Fans

- **ESP32 Gateway**

Wi-Fi TCP/HTTP Uplink to Dashboard

Detailed System Architecture

Pin-Level Interfacing and Signal Protocol Topology

SYSTEM PIN MAPPING & BUS TOPOLOGY SCHEMATIC

SENSORY TRANSDUCERS

DHT22:

GPIO (Single-Wire Data Bus)

LDR Sensor:

ADC1_IN0 (0–3.3V Analog)

PIR Motion:

EXTI_Line0 (Interrupt Trigger)

MASTER MCU CORE: STM32

ARM Cortex-M Embedded Architecture

- ADC1_IN0 (Continuous Lux Sampling)
- TIM2_CH1 (Hardware PWM Timer)
- GPIO_EXTI0 (Asynchronous Motion)
- USART2 (ESP32 Serial Uplink)
- Sub-50ms Deterministic Response

ACTUATORS & GATEWAY

Relay Module:

230V AC Classroom Lighting

MOSFET Driver:

TIM2 PWM DC Fan Speed

ESP32 Module:

USART2 -> Wi-Fi Dashboard

Hardware Components Breakdown

Technical Selection, Specifications and Operational Roles

STM32 Microcontroller

32-bit ARM Cortex-M processor delivering deterministic timer

Role: Master Core

DHT22 (AM2302)

Capacitive humidity sensor & thermistor (-40 to 80°C, ±0.5°C accuracy; 0–100% RH,

Role: Climate Monitor

LDR Sensor Module

Photoresistor with LM393 comparator providing continuous analog light lux

Role: Lux Harvesting

PIR Motion Sensor

Pyroelectric sensor with faceted Fresnel lens (HC-SR501) detecting

Role: Occupancy Latch

Optocoupled Relay

Electromechanical relay with optocoupler isolation, allowing 3.3V logic to safely switch 230V AC troffers.

Role: High-Voltage Light

MOSFET Motor Driver

Logic-level N-channel power MOSFET driven by high-frequency STM32 PWM to regulate ceiling fan speed.

Role: Fan Modulation

ESP32 Wi-Fi & BLE SoC

Dual-core 240MHz wireless microcontroller acting as communications bridge: buffers UART packets and publishes telemetry.

Role: IoT Gateway

Working Principle & Closed-Loop Cycle

Autonomous Sense-Process-Actuate Cycle

STEP 01 Sensory Polling

- DHT22 queried every 2s via bus.
- LDR sampled via 12-bit ADC.
- PIR latched via GPIO interrupt.

STEP 02 Threshold Logic

- Compare LDR to threshold ($< 40\%$).
- Compare temp to comfort brackets.
- Check occupancy timeout latch.

STEP 03 Actuator Dispatch

- Trigger GPIO high/low for relay.
- Adjust PWM register (TIM2->CCR1).
- De-energize loads if timeout.

STEP 04 Uplink Telemetry

- Assemble binary/JSON frame.
- Transmit packet over UART to ESP32.
- Publish data to web dashboard.

Deterministic Cycle Latency: Sub-50 millisecond decision loop guarantees immediate response.

Automatic Lighting Control Logic

Dual-Condition Occupancy & Daylight Harvesting Algorithm

EMBEDDED DECISION RULE

IF (Occupancy == TRUE && AmbientLight < 40%)
→ RELAY_STATE = ON (Lights Energized)

ELSE

→ RELAY_STATE = OFF (Lights Extinguished)

- **Scenario A: Occupied + Dark**

Students inside, cloudy day or evening -> LIGHTS ON

- **Scenario B: Occupied + Ample Daylight**

Students inside, bright window sunlight > 40% -> LIGHTS OFF

- **Scenario C: Vacant Classroom**

PIR inactive, zero occupants inside -> LIGHTS OFF

LIGHTING DECISION FLOW

1. Sample PIR Motion Interrupt

↓ [Human Motion Confirmed?]

2. Sample LDR Analog Lux Channel

↓ [Is Daylight < 40% Deficit?]

YES: Close Relay (Lights ON)

NO: Open Relay (Lights OFF / Energy Saved)

Automatic Fan & Thermal Regulation

Multi-Stage Temperature-to-PWM Modulation Logic

BRACKET 1

< 25.0°C

Cool Baseline

0% PWM (Fan OFF)

BRACKET 2

25.0°C – 29.9°C

Warm Mild Heat

30% PWM (Low)

BRACKET 3

30.0°C – 34.9°C

Moderate Thermal Load

60% PWM (Medium)

BRACKET 4

≥ 35.0°C

High Heat Load

100% PWM (Max)

HARDWARE PWM GENERATION

Configured via STM32 General Purpose Timer (TIM2), clocked at 10 kHz with an 8-bit duty cycle counter (0–255 steps) yielding smooth motor torque.

VACANCY CUT-OFF SAFETY

Regardless of high room temperature, if PIR reports zero occupancy for longer than the inactivity timeout, the fan automatically cuts power to 0%.

Embedded System Software Flowchart

Deterministic Main Loop Execution Model

DETERMINISTIC MAIN EXECUTION LOOP

[START / POWER ON]

↓ Init Clocks, GPIO, ADC, TIM2 PWM, USART2

↓ Sample Sensors (DHT22, LDR ADC, PIR State)

↓ < Is Classroom Occupied? >

YES: Eval Light (Lux < 40%) | Eval Fan PWM

NO: Decrement Inactivity Latch -> Cut All Loads

↓ Transmit Telemetry Packet to ESP32 over UART

↓ Deterministic Non-blocking Loop Repeat

CONTROL SAFETY FEATURES

• No Blocking Delays:

Hardware interrupts maintain continuous system responsiveness.

• Fail-Safe Design:

Sensor timeout causes actuators to default to unpowered state.

• Hardware Watchdog:

Independent Watchdog (IWDG) reboots system if loop hangs.

• Brownout Protection:

Embedded BOR ensures clean reset during voltage dips.

Software Stack & IoT Communication

STM32 Embedded Firmware & ESP32 Network Bridge

STM32 EMBEDDED FIRMWARE

Bare-Metal C / STM32CubeIDE / HAL

- **Hardware Abstraction Layer (HAL):**

Clean register abstractions for GPIO, ADC1, TIM2, and USART2 peripherals.

- **Interrupt Service Routines (ISR):**

EXTI pin trigger for immediate PIR motion event capture without CPU polling.

- **Structured Serial Packet Format:**

Serializes telemetry frames: #TEMP:28.5,HUM:55,LUX:32,PIR:1,FAN:60,LIGHT:1\$

- **Deterministic Timing:**

Strict loop scheduling with zero memory fragmentation.

ESP32 WIRELESS GATEWAY

Wi-Fi 802.11 b/g/n / FreeRTOS

- **Hardware UART Ring Buffer:**

Reads STM32 packet stream via UART RX/TX without loading the main STM32 core.

- **Campus Wi-Fi Stack:**

Establishes connection to institutional Wi-Fi access points using WPA2-Enterprise / WPA2-PSK.

- **Telemetry Serving:**

Hosts asynchronous WebSocket or HTTP REST endpoints for real-time monitoring and manual override commands.

- **Decoupled Reliability:**

STM32 control continues operating safely even if Wi-Fi disconnects.

Expected Operation & Test Scenarios

System Response Under Realistic Field Conditions

Scenario 01: Classroom Vacant at Morning (7:00 AM) → Lights: OFF | Fan: 0% (Idle Baseline)

PIR reports 0 occupancy, external lux 20%, room temp 22°C.

Scenario 02: Students Enter on Rainy Morning (8:00 AM) → Lights: ON (Auto Relay) | Fan: 0%

PIR triggers active, external lux drops to 15% (insufficient daylight).

Scenario 03: Sunny Afternoon Lecture (1:00 PM) → Lights: OFF (Daylight Harvest) | Fan: 60% PWM

Classroom fully occupied, outdoor solar lux surges to 85%, room temp hits 32°C.

Scenario 04: Summer Heat Wave Peak (3:00 PM) → Lights: Evaluated | Fan: 100% PWM Max Airflow

Classroom occupied, thermal load surges to 37.5°C.

Scenario 05: Class Dismissal / Evening Exit (5:00 PM) → Zero Waste Shutdown: All Actuators OFF

Students exit through doorway; PIR latches clear; inactivity countdown expires.

Advantages & Future Expansion

Quantifiable Engineering Benefits and Next-Phase Roadmap

CORE ADVANTAGES

✓ Up to 40% Energy Reduction:

Eliminates human error, parasitic overnight loads, and unnecessary midday artificial lighting troff

✓ Autonomous Student Comfort:

Continuous proportional cooling prevents classroom thermal fatigue during intense lectures.

✓ Decoupled Architecture:

STM32 ensures rock-solid deterministic safety even if Wi-Fi or cloud networks disconnect.

✓ Low Retrofit Cost:

Compatible with standard legacy 230V fluorescent/LED arrays and common ceiling fan installati

FUTURE EXPANSION ROADMAP

➔ Campus-Wide Mesh Deployment:

Implement ESP-NOW or LoRaWAN mesh networking across hundreds of university lecture halls and labs

➔ Predictive Pre-Cooling with ML:

Integrate academic timetable databases to automatically pre-condition rooms 5 minutes before scheduled

➔ Hardware Current Shunt Monitoring:

Integrate INA219 current sensors to log exact real-time kilowatt-hour power consumption curves.

➔ Motorized Blinds Integration:

Modulate window blinds based on sun glare angle to further lower thermal absorption.

Conclusion

Project Summary & Engineering Value Proposition

PROJECT SUMMARY

The proposed Smart Classroom Automation System Using STM32 delivers an integrated, cost-effective, and highly reliable embedded solution to combat energy waste in academia.

- **Multi-Sensory Closed-Loop Control:**

Combines ambient lux (LDR), passive motion (PIR), and temperature (DHT22) for accurate decisions.

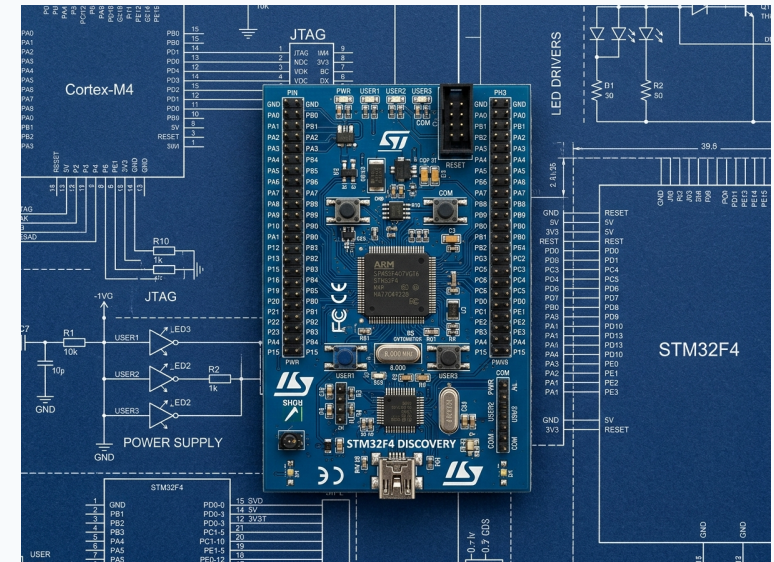
- **Energy & Comfort Balance:**

Eliminates daylight redundancy and empty classroom waste while maintaining adaptive proportional ventilation.

- **Dual-Layer IoT Scalability:**

Robust STM32 hardware controller paired with ESP32 Wi-Fi gateway for centralized oversight.

"Smarter Classroom, Better Comfort, Efficient Energy Use"



THANK YOU

Questions, Discussions & Evaluator Feedback

Project: Smart Classroom Automation System Using STM32

Department of Electronics & Communication / Embedded Systems Engineering

Interactive 3D Digital Twin Simulation Available for Live Evaluation Demonstration